- 1/3 with symptoms have COVID-19
- Let C be the even that a person has COVID-19
- Let E be the event that a person has tested positive for COVID-19

$P(C) = 1/3 = 0.333\ldots$ $\quad p(C^C) = 1-(1/3) = 0.666\ldots$

$P(E|C) = 0.985$ $\quad p(E^C|C) = 1 - 0.985 = 0.015$

$P(E|C^C) = 1-0.8 = 0.2$ $\quad p(E^C|C^C) = 0.8$

1. $P(C|E) = \frac{p(E|C)p(C)}{p(E|C)p(C)+p(E|Cc)p(Cc)}$

   $= \frac{(0.985)(\frac{1}{3})}{(0.985)(\frac{1}{3})+(0.2)(2/3)}$

   $= \frac{0.328333\ldots}{0.461666\ldots}$

   ~ 0.711191335

   $\therefore$ The probability that a randomly selected person showing symptoms has COVID-19 if the test returns positive is ~0.711

   $\blacksquare$

2. $P(C^C|E^C) = \frac{p(Ec|Cc)p(Cc)}{p(Ec|Cc)p(Cc)+p(Ec|C)p(C)}$

   $= \frac{(0.8)(\frac{2}{3})}{(0.8)(\frac{2}{3})+(0.015)(1/3)}$

   $= \frac{0.53333\ldots}{0.538333\ldots}$

   ~ 0.990712073

   $\therefore$ The probability that a randomly selected person showing symptoms does not have COVID-19 if the test returns negative is ~0.991

   $\blacksquare$